

Task-3

Impact of Input Data Normalization Methods on the Results of TOPSIS

Submission format: source files (e.g., notebook (.ipynb) or .py, etc.) + report (PDF, approximately few pages)

1. Educational objective

The goal of this laboratory is to analyze how the choice of normalization method changes the final ranking produced by the TOPSIS method.

Students should learn how to:

- apply several normalization methods to the same decision matrix,
- perform TOPSIS calculations on consistently defined criteria,
- compare rankings obtained under different normalization assumptions,
- interpret numerical differences, not only final ranks,
- formulate conclusions about the sensitivity of TOPSIS to preprocessing.

Key point: the report should show whether normalization has a **real impact** on the decision result, not only present repeated calculations.

2. Theoretical background

2.1. TOPSIS

TOPSIS (Technique for Order Preference by Similarity to Ideal Solution) is a multi-criteria decision-making method in which the best alternative should be as close as possible to the ideal solution and as far as possible from the anti-ideal solution.

For a decision matrix

$$X = [x_{ij}], \quad i = 1, 2, \dots, n, \quad j = 1, 2, \dots, m,$$

where n is the number of alternatives and m is the number of criteria, the method requires normalization before weighting and distance calculations.

2.2. Why normalization matters

Criteria are often expressed in different units and scales, for example cost, time, quality, energy consumption, or customer rating. Without normalization, criteria with numerically larger values may dominate the result. Different normalization methods transform the original data in different ways, which may change:

- the relative distances between alternatives,
- the balance between criteria,
- the final TOPSIS scores,
- the ranking order.

3. Task

Use the **same dataset, criteria, preference directions (profit/cost), and weights** as in the previous TOPSIS laboratory.

Then apply **five different normalization methods** and compare the obtained results.

3.1. Normalization methods to be tested

1. Vector normalization

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^n x_{ij}^2}}$$

2. Min-max normalization

$$r_{ij} = \frac{x_{ij} - \min_i x_{ij}}{\max_i x_{ij} - \min_i x_{ij}}$$

3. Max normalization

$$r_{ij} = \frac{x_{ij}}{\max_i x_{ij}}$$

4. Sum normalization

$$r_{ij} = \frac{x_{ij}}{\sum_{i=1}^n x_{ij}}$$

5. Z-score standardization

$$r_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

3.2. Explanation of symbols

- x_{ij} – original value of alternative i under criterion j ,
- r_{ij} – normalized value of alternative i under criterion j ,
- n – number of alternatives,
- m – number of criteria,
- $\max_i x_{ij}$ – maximum value in criterion j ,
- $\min_i x_{ij}$ – minimum value in criterion j ,
- μ_j – arithmetic mean for criterion j ,
- σ_j – standard deviation for criterion j .

4. TOPSIS procedure

For each normalization method, perform the full TOPSIS procedure.

4.1. Step 1: Build the normalized matrix

Create the matrix

$$R = [r_{ij}].$$

4.2. Step 2: Apply weights

Construct the weighted normalized matrix:

$$v_{ij} = w_j r_{ij},$$

where w_j is the weight of criterion j and

$$\sum_{j=1}^m w_j = 1.$$

4.3. Step 3: Determine the ideal and anti-ideal solutions

For profit criteria:

$$A^+ = \left\{ \max_i v_{ij} \right\}, \quad A^- = \left\{ \min_i v_{ij} \right\}$$

For cost criteria:

$$A^+ = \left\{ \min_i v_{ij} \right\}, \quad A^- = \left\{ \max_i v_{ij} \right\}$$

where A^+ is the ideal solution and A^- is the anti-ideal solution.

4.4. Step 4: Compute distances

For each alternative i , calculate the Euclidean distances:

$$S_i^+ = \sqrt{\sum_{j=1}^m (v_{ij} - A_j^+)^2}, \quad S_i^- = \sqrt{\sum_{j=1}^m (v_{ij} - A_j^-)^2}$$

where S_i^+ is the distance from the ideal solution and S_i^- is the distance from the anti-ideal solution.

4.5. Step 5: Compute the relative closeness coefficient

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}$$

where $C_i \in [0, 1]$. A larger value of C_i means a better alternative.

5. Required analysis

The report should not stop at presenting five rankings. It should contain a clear comparison and interpretation.

5.1. Compare the results

For each normalization method, present:

- the final TOPSIS score C_i for each alternative,
- the final ranking,
- the position of the best and worst alternative.

Then compare the methods and answer the following questions:

- Does the ranking change across normalization methods?
- Which alternatives change position most strongly?
- Are the score differences small or substantial?
- Does the winning alternative remain the same?
- Which normalization method appears the most stable?

5.2. Creative interpretation

This laboratory requires independent thinking. Useful ideas include:

- comparing rank shifts between methods,
- measuring the difference between the highest and lowest TOPSIS score for the same alternative,
- identifying whether a method compresses or amplifies differences,
- discussing sensitivity to outliers or narrow data ranges,
- explaining how preprocessing may influence a practical decision.

6. Structure of the report

The report should include:

1. Title page: course, topic, student information
2. Short description of the original decision problem
3. Decision matrix, criteria types, and weights
4. Description of the 5 normalization methods
5. TOPSIS results for each method
6. Comparative analysis with tables or plots
7. Final conclusions

7. Conclusion

In the final section, explain whether normalization significantly affects the TOPSIS result.

The conclusion should clearly answer:

- whether normalization changes the ranking,
- when the choice of normalization becomes critical,
- which method seems the most reasonable for your dataset,
- what practical lesson follows from the experiment.

Important note: originality, quality of comparison, and depth of reflection are important. A correct solution should include both computations and interpretation.